Case 1: A situation in which the relationships between interest, money, and time do not exist

!!!! When choosing between different materials, one should not consider only the price of the material; factors such as the type of material and transportation costs—which can also affect overall expenses—must also be taken into account.

A company uses **stainless steel, priced at 12 TL/kg**, to manufacture bolts. Management wants to know whether the same bolts could be produced at a lower cost if **brass—priced at 18 TL/kg—**were used instead.

The annual market demand for bolts is 150,000 units. The type of material used is not a significant factor regarding market demand.

The amount of **steel required for one bolt is 0.0373 kg**, while the amount of **brass required is 0.0384 kg.**

The production rate is **60 bolts per hour when steel is used**, and **115 bolts per hour when brass is used.**

The wage for the machine operator is 10 TL/hour, and **the general manufacturing costs for the machine are 8 TL/hour.**

Answer:

Cost Type	Steel	Brass
Material	12 tl x 0.0373 kg =0.4476 tl	18 tl x 0.0384 kg=0.6912 tl
Labour	10tl / 60 units = 0.1666 tl	10tl /115 units = 0.0870 tl
Overheads	8 tl / 60 units = 0.1333 tl	8 tl / 115 units = 0.0696 tl
Total cost for bolts	= 0.7475 tl	= 0.8478

Savings per unit when steel is used:

0.8478 – 0.7475 = 0.1003 tl

Since the annual production volume is 150,000 units, the revenue remains constant across the alternatives. Due to the difference in material costs, using stainless steel for bolt manufacturing proves more advantageous, resulting in an annual saving of 15,045 TL.
(150000 units x 0.1003 = 15.045 tl)